Pythagorean Theorem

THE PYTHAGOREAN THEOREM IS USED TO FIND THE LENGTH OF MISSING SIDES IN RIGHT TRIANGLES.

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If the lengths of two sides of a right triangle are known, the length of the third side can be worked out using Pythagorean Theorem.

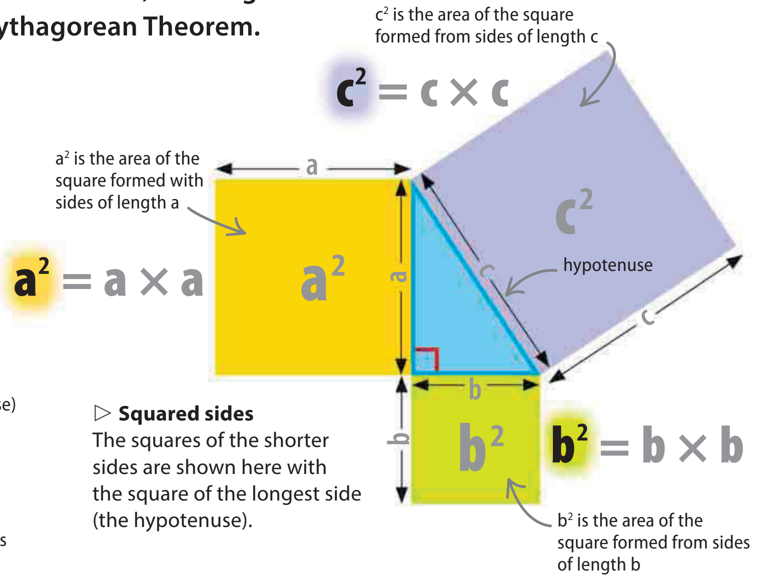
What is the Pythagorean Theorem?

The basic principle of the Pythagorean Theorem is that squaring the two smaller sides of a right triangle (multiplying each side by itself) and adding the results will equal the square of the longest side. The idea of "squaring" each side can be shown literally. On the right, a square on each side shows how the biggest square has the same area as the other two squares put together.

side a, side b, side c (hypotenuse)

$$a^2 + b^2 = c^2$$

formula shows that side a squared plus side b squared equals side c squared



▷ **Squared sides**
The squares of the shorter sides are shown here with the square of the longest side (the hypotenuse).

If the formula is used with values substituted for the sides a, b, and c, the Pythagorean Theorem can be shown to be true. Here the length of c (the hypotenuse) is 5, while the lengths of a and b are 4 and 3.

a is 4, b is 3, c is 5

$$a^2 + b^2 = c^2$$

$$4^2 + 3^2 = 5^2$$

4×4, 3×3, 5×5

$$16 + 9 = 25$$

squares of two shorter sides added; square of hypotenuse

△ **Pythagoras in action**
In the equation the squares of the two shorter sides (4 and 3) added together equal the square of the hypotenuse (5), proving that the Pythagorean Theorem works.